Technical University of Munich // TUM University Hospital
TUM School of Medicine and Health // TUM School of Computation, Information, and Technology
Munich Institute of Robotics and Machine Intelligence (MIRMI)

Computer-Assisted Interventions & Therapy

Introduction

Prof. Dr. Daniel Roth, Prof. Nassir Navab, PhD

Technical University of Munich

TUM School of Medicine and Health, TUM School of Computation, Information, and Technology

13.10.2026

Agenda

1. Introduction of hosts and teams
2. What is computer-assisted interventions & therapy?
3. Course dates, independent projects, & assessments
4. Project exposés and questions
5. Project preference voting
6. Next steps & closing

Hosts and team

Prof. Daniel Roth
Human-Centered Computing and Extended Reality (HEX)

Human-Centered Computing and Extended Reality Lab



Prof. Dr. Daniel Roth

- **TUM**
- **TUM Klinikum**
- **Munich Institute of Robotics and Machine Intelligence (MIRMI)**



Orthopedics & Sports Orthopedics

- Klinikum rechts der Isar
- Clinic for Orthopedics and Sports Orthopedics
  Director: Prof. Rüdiger von Eisenhart-Rothe



Dr. Julian Kreimeier

Computer Vision | **Robotics** | **Extended Reality** | **Simulation** | **AI**



Hannah Schieber



Shiyu Li



Victor Schaack



Kristoffer Waldow



Constantin Kleinbeck



Judith Friedl-Knirsch



Luisa Theelke



Doruk Bildibay



Elisabeth Schiele



Abdelrahman Elskhawy



Dimash Mukashev



Daniel Homm

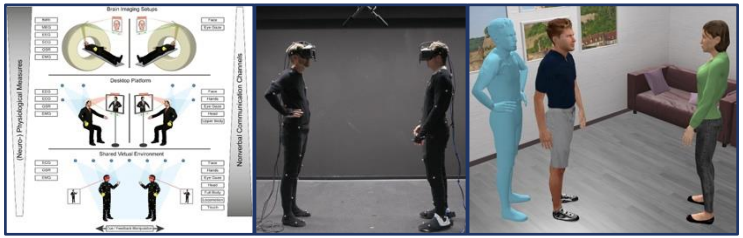
RESEARCH
Human-Centered Computing and Extended Reality

PEOPLE
Human-Centered Computing and Extended Reality

Selected Awards

- **2025 IEEE VR Best Paper Honorable Mention**
- **2024 IPCAI - Bench to Bedside Award** (1/1) *Int. Conf. on Information Processing in Computer-Assisted Interventions*
- **2024 IEEE VR - Best Poster Award** (1/5) *IEEE Virtual Reality Conference*
- **2024 CVPR Highlight Paper** (3%) *IEEE/CVF Conference on Computer Vision and Pattern Recognition*
- **2022 MARSS Best Project Award in Medical Education** (1/1) *Medical Augmented Reality Summerschool*
- **2021 IEEE VR Best Demo Honorable Mention Award** (1/1) *IEEE Virtual Reality and 3D User Interfaces*
- **2020 ISMAR Best Poster Award** (1/1) *IEEE International Symposium on Mixed and Augmented Reality*
- **2020 Best Paper Nominee** (1/4) *IEEE International Conference on Artificial Intelligence & Virtual Reality*
- **2018 Best Journal Paper Award** (1/1) *Transactions on Visualization and Computer Graphics(IEEE VR Special Issue)*

Imagineering novel languages for the communication between human(s) and machine(s).



ACM VRST 2019



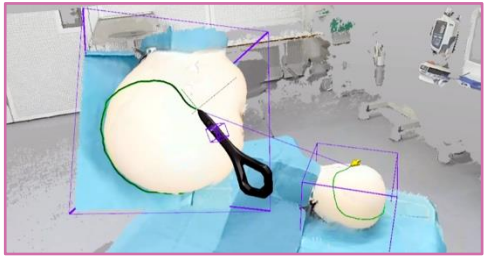
IEEE VR 2019



IEEE VR 2024



IEEE ISMAR 2023



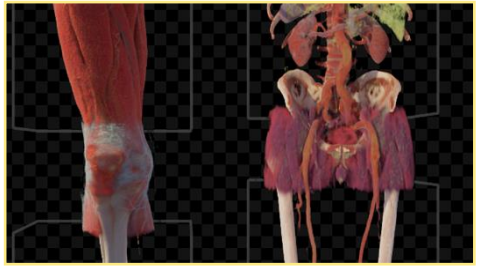
IEEE VR 2021



IPCAI 2024



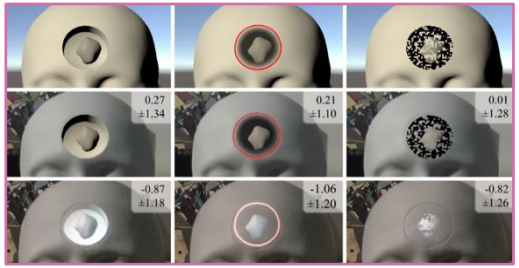
IEEE VR 2021



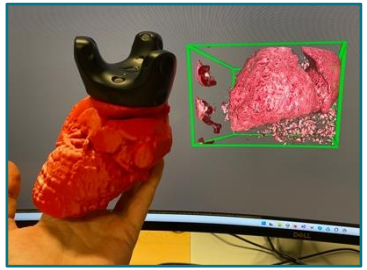
IEEE VR & TVCG 2025



IEEE ISMAR 2020

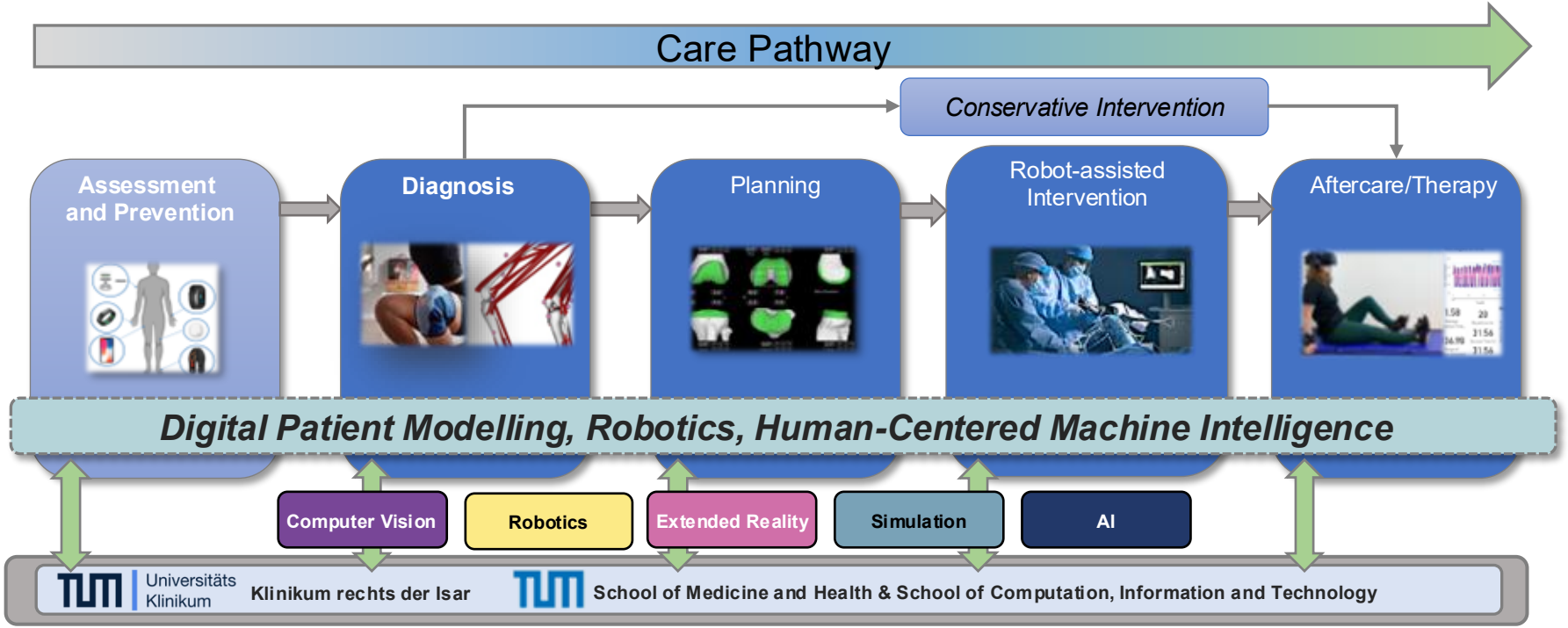


IEEE ISMAR & TVCG 2021



IEEE ISMAR 2023

Human-Centered Computing and Extended Reality Lab

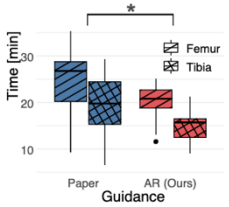


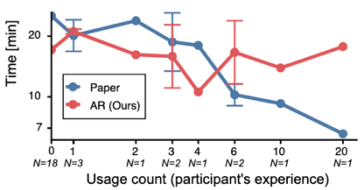
Prof. Dr. Daniel Roth (TUM) | daniel.roth@tum.de

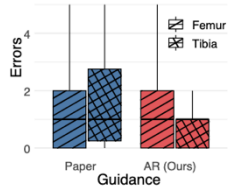
Human-Centered Computing and Extended Reality

Instrument Handling Support & Training



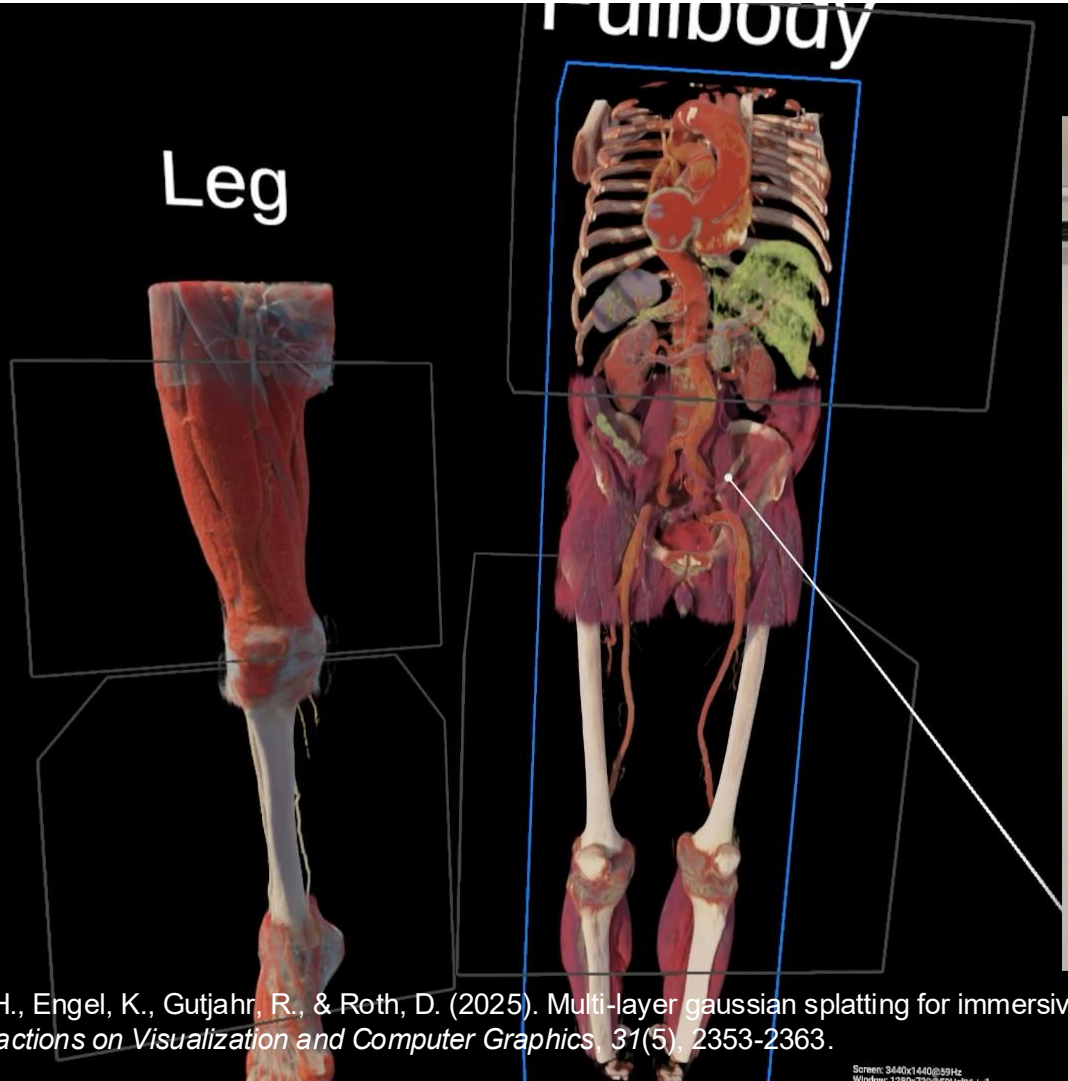






	Paper			AR (ours)			Sign	Effect
	M	MD	SD	M	MD	SD	$p \leq$	r/d
Raw NASA-TLX								
Mental	75.862	80.0	17.882	43.276	40.0	26.332	.001	$r = 0.08$
Physical	35.345	20.0	27.611	27.241	20.0	23.776	-	-
Temporal	73.276	80.0	20.627	33.448	25.0	24.279	.001	$r = 0.96$
Performance	53.448	55.0	24.389	30.172	30.0	18.540	.001	$d = 1.07$
Effort	63.450	70	22.955	32,07	30.0	15.150	.001	$d = 1.59$
Frustration	68.276	70.0	20.366	33.966	30.0	23.160	.001	$d = 1.57$
User Experience Questionnaire (UEQ)								
Attractiveness	-0.851	-0.83	1.306	1.925	2.0	0.921	.001	$r = 1.06$
Perspicuity	-0.940	-1.5	1.304	1.603	1.75	1.030	.001	$r = 1.02$
Efficacy	-1.034	-1.0	1.166	1.276	1.0	0.974	.001	$r = 1.03$
Dependability	-0.069	-0.5	1.318	-1.543	1.5	0.906	.001	$d = -1.43$
Stimulation	-0.310	-0.5	1.260	2.320	2.5	0.759	.001	$r = 1.05$
Novelty	-1.457	-1.75	1.379	2.034	2.5	1.085	.001	$r = 1.11$
SUS	43.966	42.500	18.207	73.362	80.000	17.299	.001	$r = 0.92$

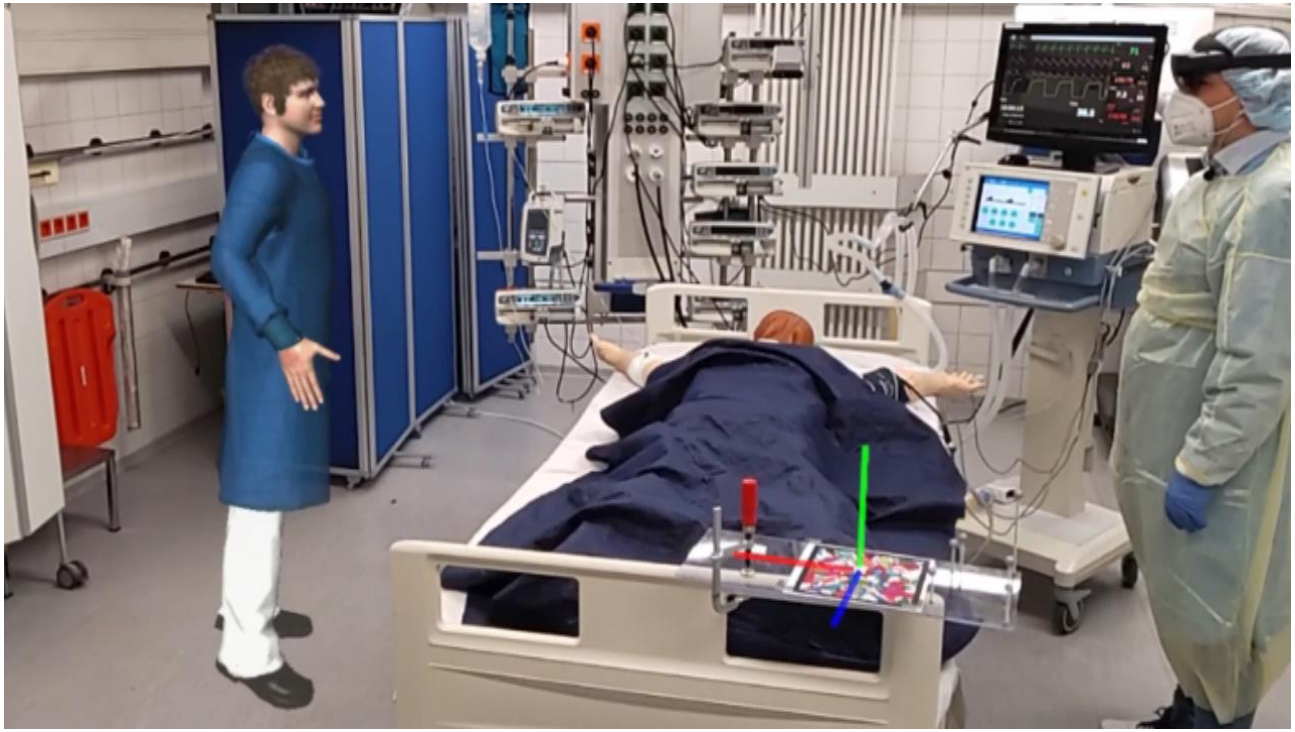
Kreimeier J., Li, S., Schieber, H., Müller, D., von Eisenhart-Rothe, R., Kainz, B. Roth,D. Design and Evaluation of Augmented Reality Guidance for Scrub Nurses' Instrumentation Guidance in Revision Arthroplasty. [under review]





Kleinbeck, C., Schieber, H., Engel, K., Gutjahr, R., & Roth, D. (2025). Multi-layer gaussian splatting for immersive anatomy visualization. *IEEE Transactions on Visualization and Computer Graphics*, *31*(5), 2353-2363.

Cross-Reality Teleconsultation





Roth, D., Yu, K., Pankratz, F., Gorbachev, G., Keller, A., Lazarovici, M., ... & Eck, U. (2021, March). Real-time mixed reality teleconsultation for intensive care units in pandemic situations. In *2021 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)* (pp. 693-694). IEEE.
Yu, K., Winkler, A., Pankratz, F., Lazarovici, M., Wilhelm, D., Eck, U., ... & Navab, N. (2021, March). Magnoramas: Magnifying Dioramas for Precise Annotations in Asymmetric 3D Teleconsultation. In *2021 IEEE Virtual Reality and 3D User Interfaces (VR)* (pp. 392-401). IEEE

Drone-based Emergency Telemedicine





Emergency situation difficult to access



Drone deployment from mobile hub

Rescue arrival, drone deployment



Autonomous scene exploration, SLAM

Operator can view the captured scenario in 3D



Gaussian Splatting, sparse mesh, ...

Operator can annotate to guide ground personnel

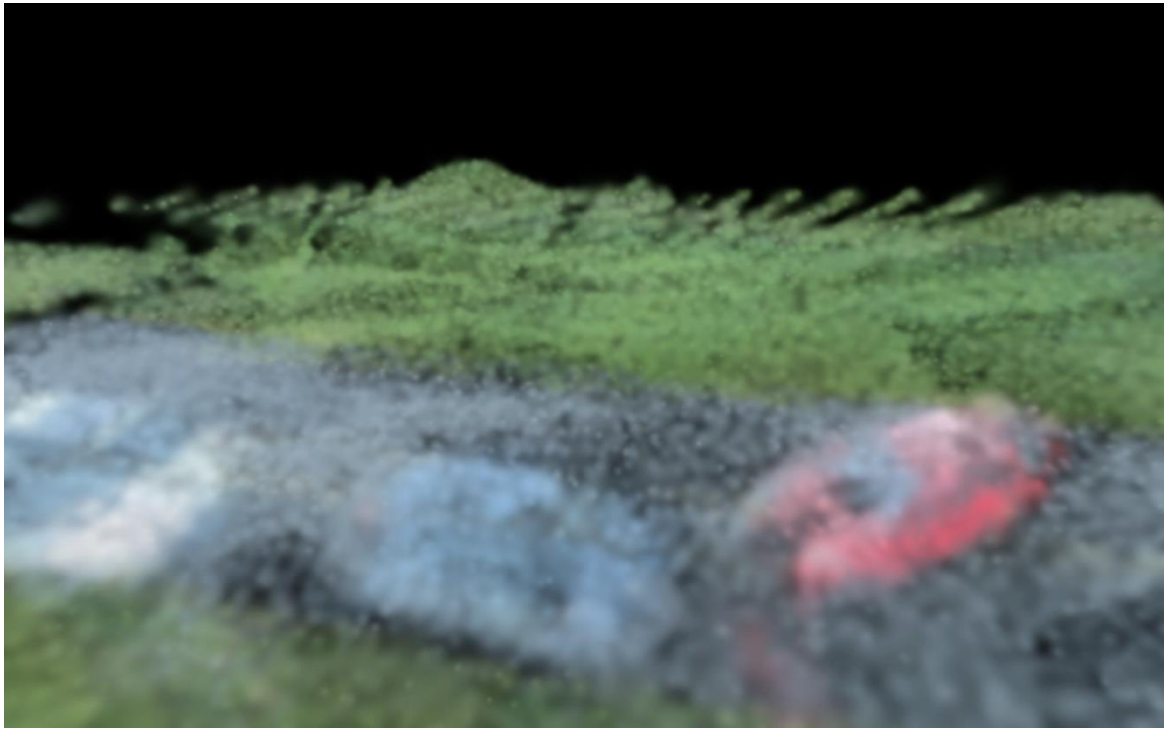


Immersive XR visualization and drone control

Schieber, H., Frischmann, D., Schaack, V., Boche, S., Schoellig, A., Leutenegger, S., & Roth, D. (2025). CoRe-GS: Coarse-to-Refined Gaussian Splatting with Semantic Object Focus. *arXiv preprint arXiv:2509.04859*.

Drone-based Emergency Telemedicine





Schieber et al., in prep.

Prof. Nassir Navab
Computer Aided Medical Procedures and Augmented Reality (CAMP)

CAMP Introduction

Over 240 MICCAI papers, Over 120 CVPR papers, Over 60 awards, …



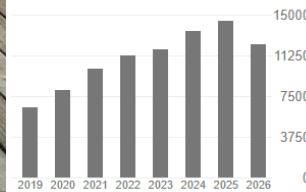




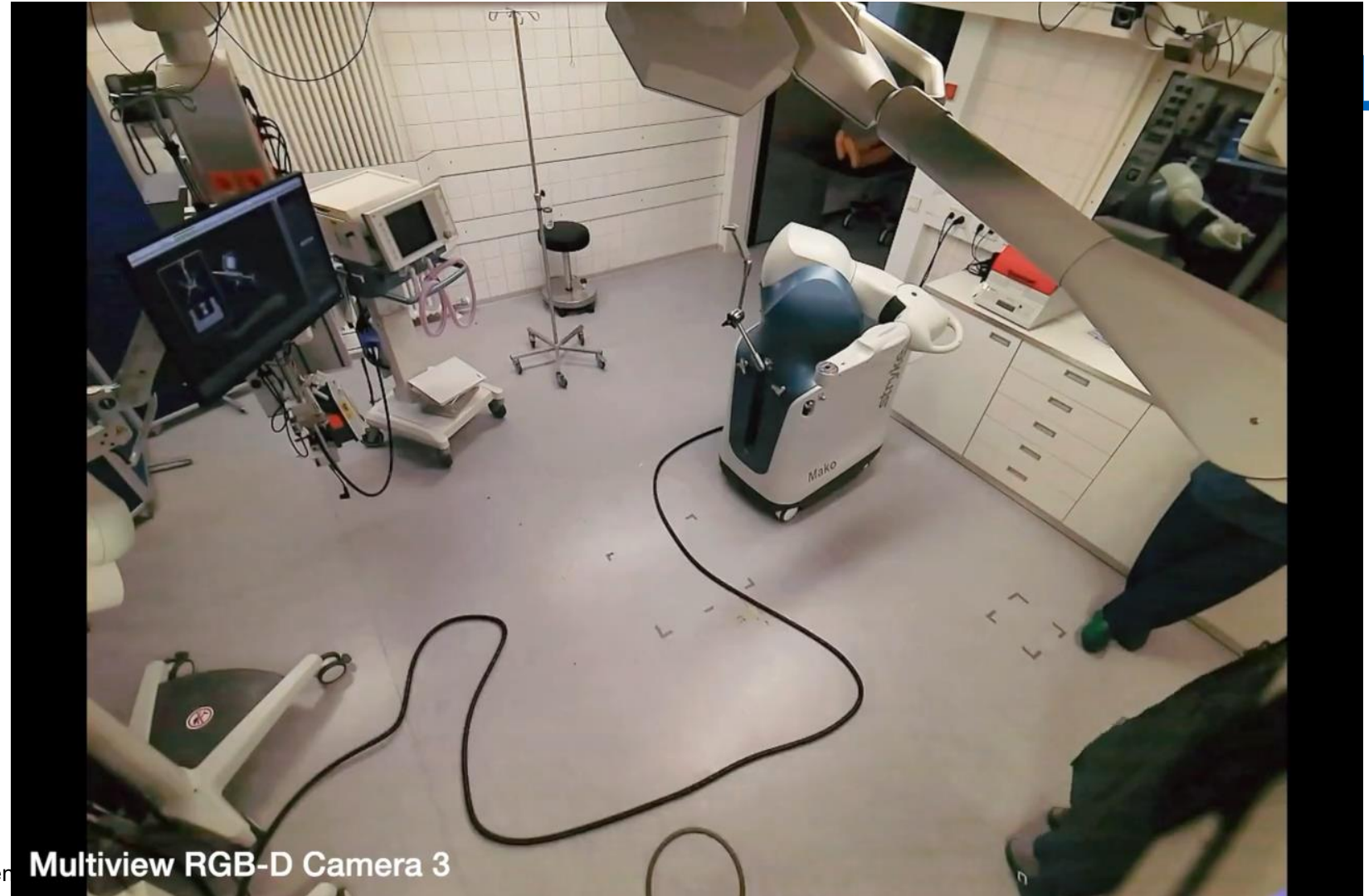




	All	Since 2021
Citations	113436	73646
h-index	139	106
i10-index	1136	797



N. Navab,- Computer Aided Medical Procedures
- Technical University of Munich


Mako
Multiview RGB-D Camera 3

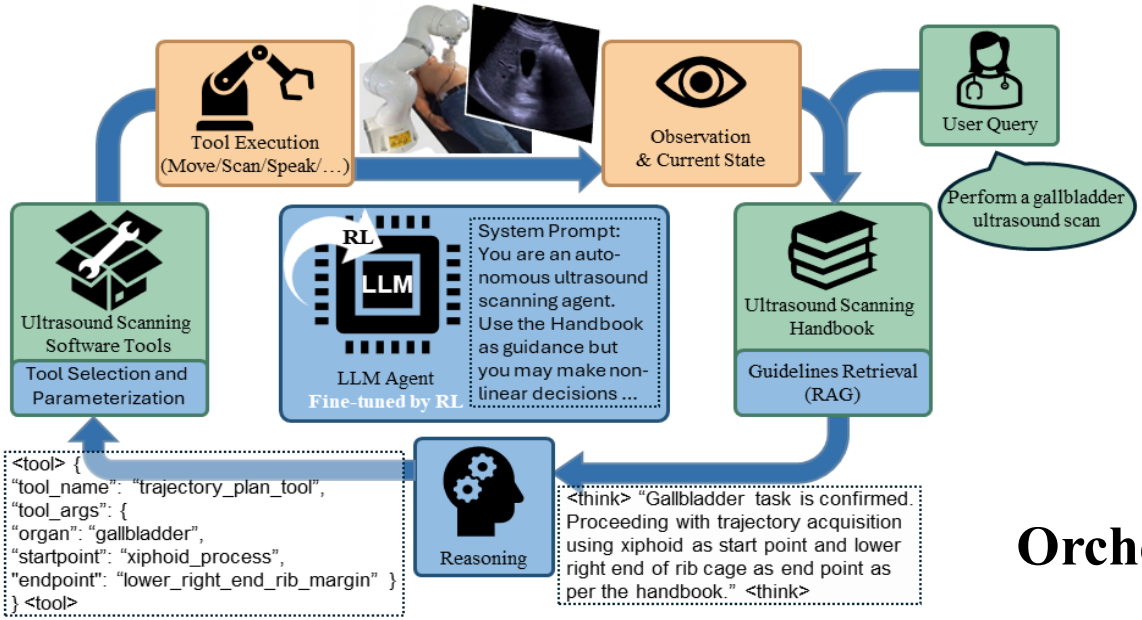
Live demonstration at Hamlyn-MRC Day





Bi, Y.*, Zhou, Y.*, Liu, P., Li, F., Jiang, Z. and Navab, N., 2026. From Scanning Guidelines to Action: A Robotic Ultrasound Agent with LLM-Based Reasoning. MICCAI 2026

LLM Orchestrator for RUSS



Orchestrator

Bi, Y.*, Zhou, Y.*, Liu, P., Li, F., Jiang, Z. and Navab, N., 2026. From Scanning Guidelines to Action: A Robotic Ultrasound Agent with LLM-Based Reasoning. MICCAI 2026

LLM Orchestrator for RUSS

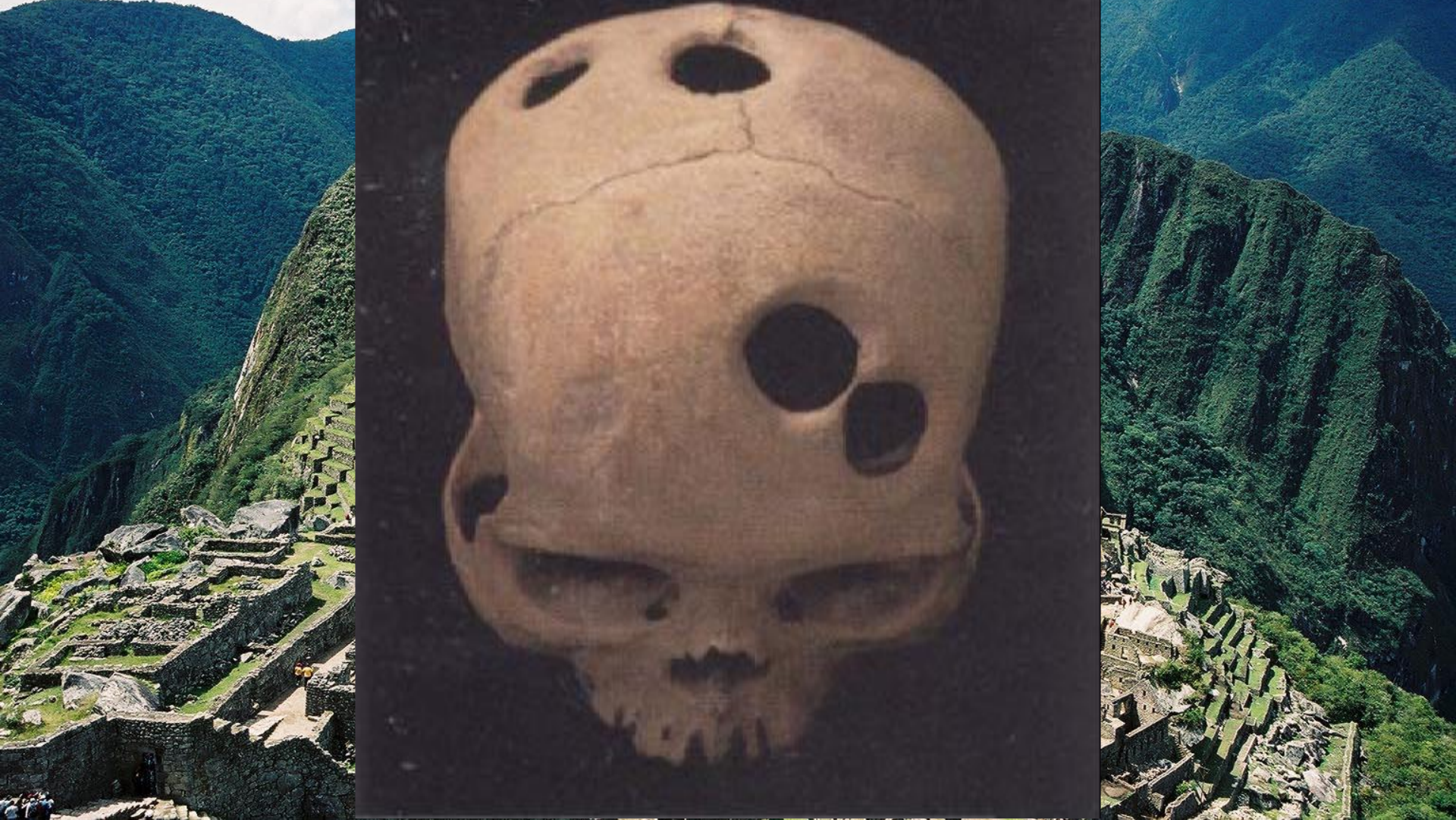


Reasoning:
The scan result has been verified after path adjustment, the repeat scan executed successfully, and the robot is at the end of the scan path. The robot should now return to the capture pose to conclude the procedure.

Calling Reset Home Tool

Bi, Y.*, Zhou, Y.*, Liu, P., Li, F., Jiang, Z. and Navab, N., 2026. From Scanning Guidelines to Action: A Robotic Ultrasound Agent with LLM-Based Reasoning. MICCAI 2026

A Historical Perspective



Ancient Greece (as early as 400BC)



1. Greek surgeons were experts in treating dislocated joints and fractures.

2. Greek surgeons did not attempt operations on the inner part of the body.

3. Later, the Romans developed sophisticated tools to remove goitres from the neck and polyps from the nose.

Operating room, early last century!



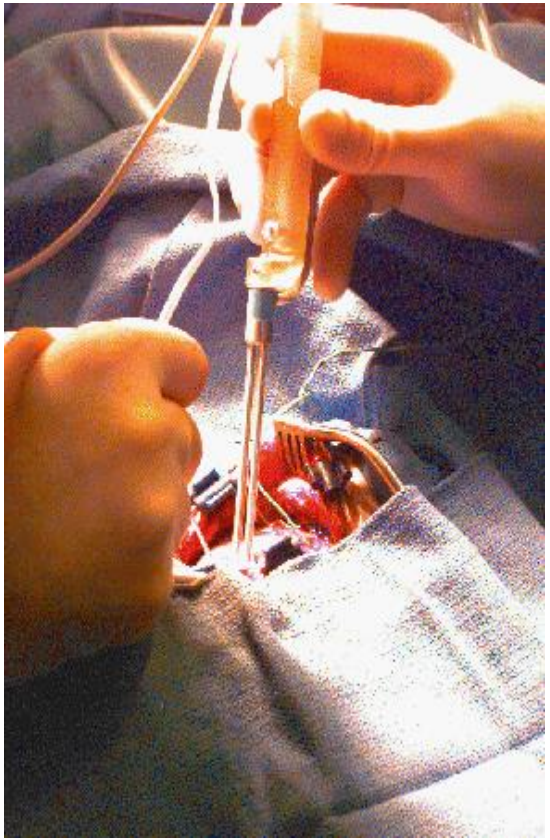
Video provided thanks to Prof. Hubertus Feussner, MITI, MRI, TUM

Operating room, mid last century!



Last Century: See, then cut

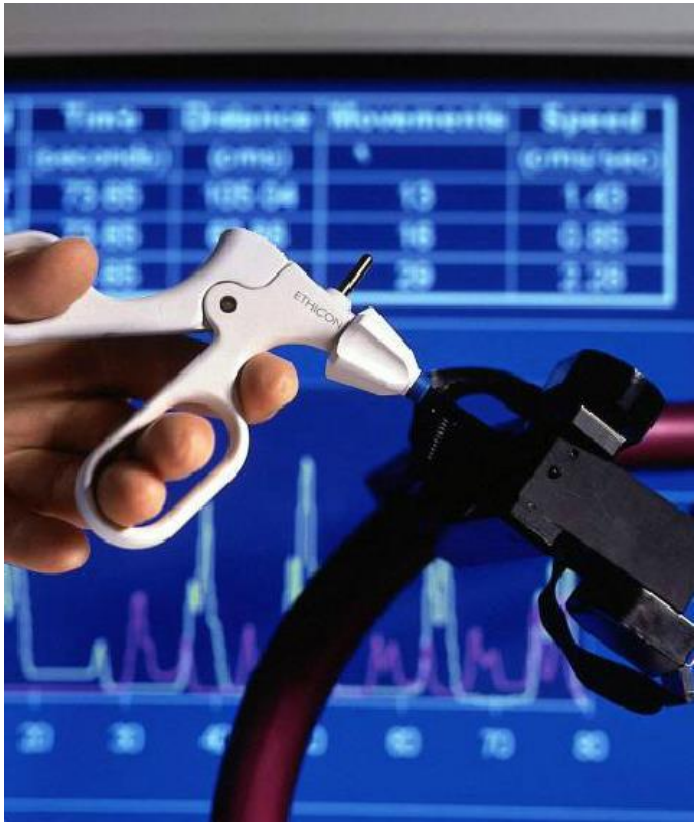




Preoperative Imaging

Intra-op execution

Laparoscopic Surgery 1980s



1. Also termed Minimally Invasive Surgery.

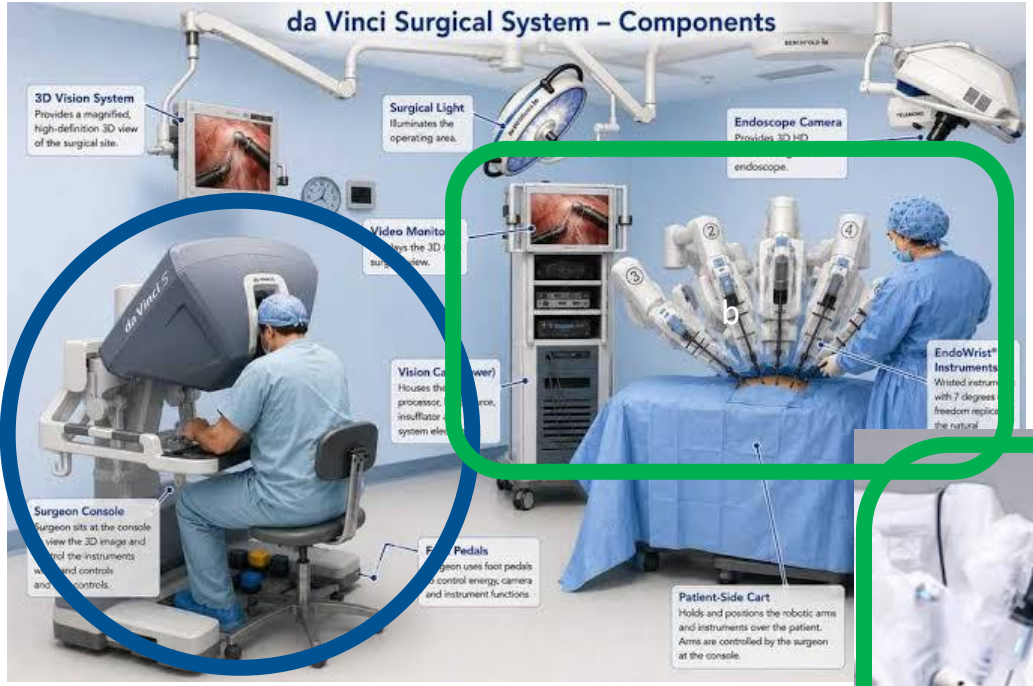
2. Operations in the abdomen are performed through small incisions.

3. In 1982, introduction of the computer chip TV camera provided the means to project operating view on a monitor.

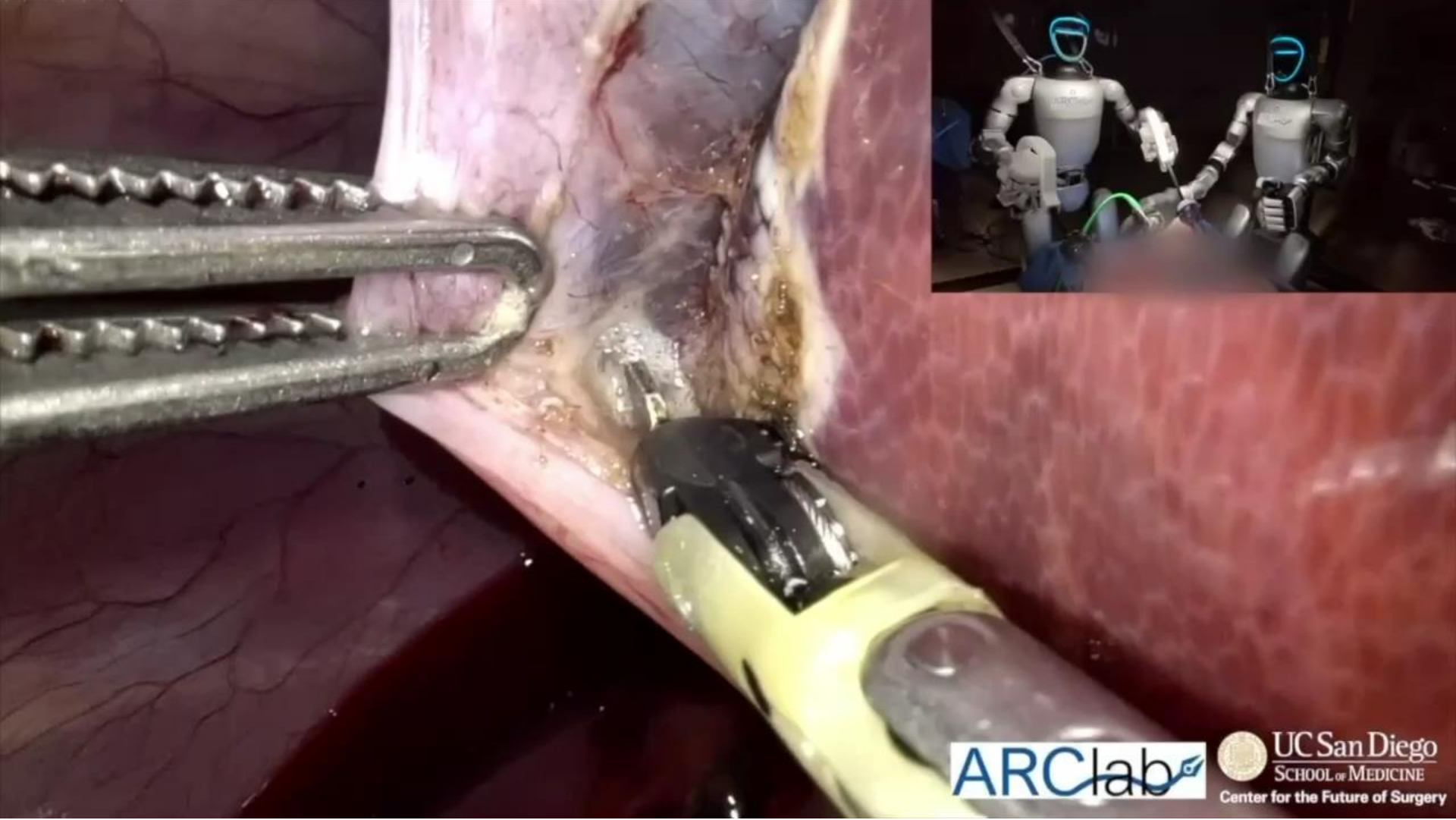
MIRS – Intuitive Surgical Inc. – da Vinci Surgical System (2001)






da Vinci Surgical System – Components
3D Vision System
Provides a magnified, high-definition 3D view of the surgical site.
Surgical Light
Illuminates the operating area.
Endoscope Camera
endoscope.
Surgeon Console
Patient-Side Cart
Holds and positions the robotic arms and instruments over the patient. Arms are controlled by the surgeon at the console.
EndoWrist® Instruments
b


b


ARClab
UC San Diego
School of Medicine
Center for the Future of Surgery



Prof. Oh Namkee

HOME > HOSPITAL

[On the Scene] 1 surgeon, 2 humanoid robots: Samsung tests the future of the operating room

Add KBR as a preferred source on Google

Kim Ji-hye | Published 2026.09.17 16:52

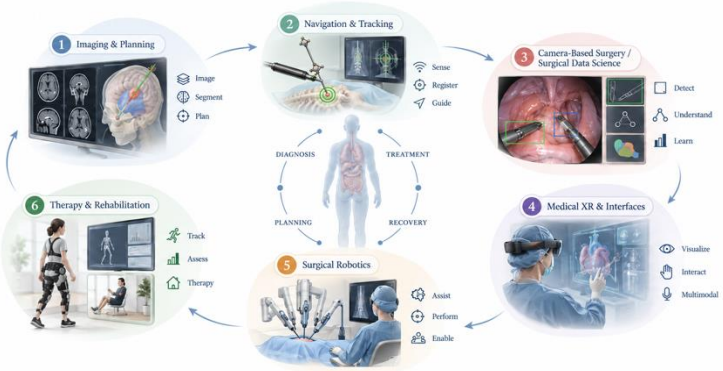
SRT-H: A hierarchical framework towards autonomous surgery



SRT-H: A hierarchical framework for autonomous surgery via language-conditioned imitation learning, Ji Woong Kim, Juo-Tung Chen, Pascal Hansen, Lucy Xiaoyang Shi, Antony Goldenberg, Samuel Schmidgall, Paul Maria Scheikl, Anton Deguet, Brandon M White, De Ru Tsai, Richard Jaepyeong Cha, Jeffrey Jopling, Chelsea Finn, Axel Krieger, Science robotics 10 (104), eadt5254

What is computer-assisted interventions & therapy?

What do we mean by CAIT?



INTERVENTIONS

Assist clinical procedures

Systems that help clinicians plan, perceive, navigate, guide, or execute a targeted medical intervention.

THERAPY

Support treatment over time

Systems that measure progress, provide feedback, personalize exercises, or assist recovery.

Closing the loop between observation, understanding, and action.

The person receiving care and the clinician remain central.

A surgeon operating. What information would actually help — and how can a computer supply it safely?



WHAT WE CAN OBSERVE

Sensors & data

Video, medical images, instruments, physiological measurements

WHAT WE NEED TO INFER

Context & intent

Anatomy, procedure phase, tool motion, risk and uncertainty

WHAT WE WANT TO IMPROVE

Decisions & actions

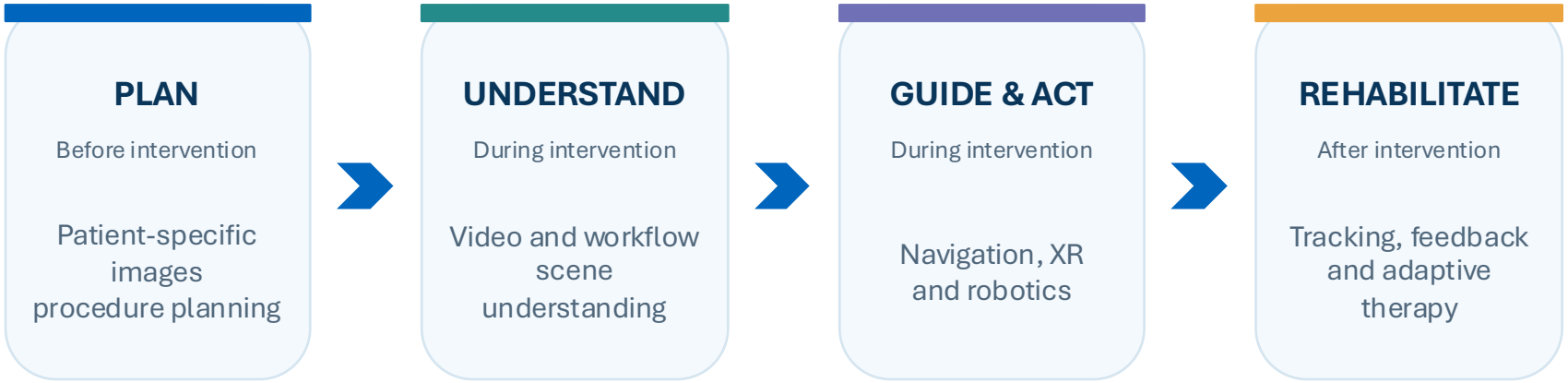
Guidance, human–robot cooperation, safer execution

Technical performance alone is not sufficient: the system must work inside a real clinical workflow.

The right information to the right person at the right time.

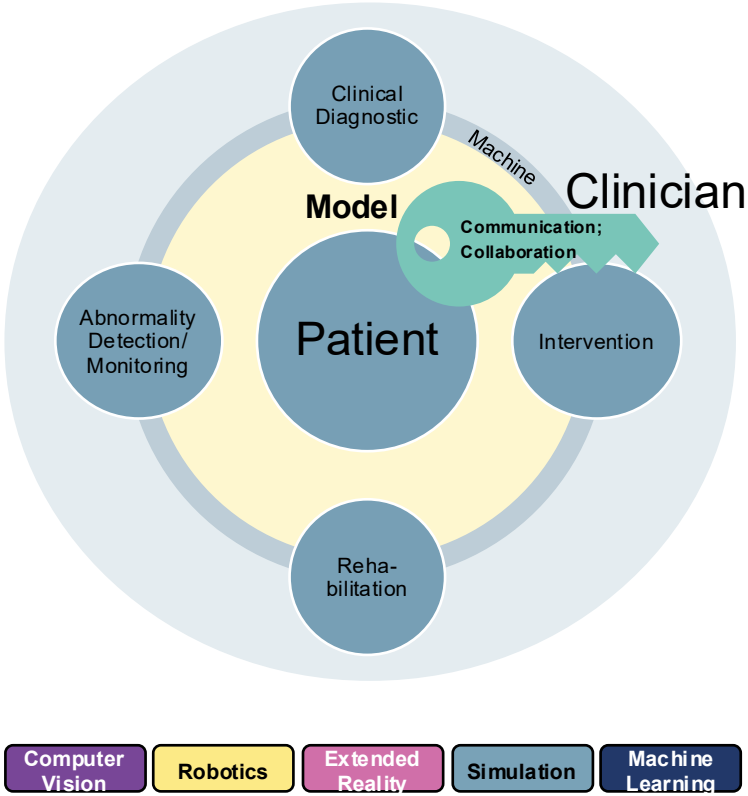
Human-Centered Computing and Extended Reality

The Patient-Care Pathway is the Shared Canvas



CAIT can support a chain of decisions, devices, data and people.

Care Transformation through CAIT

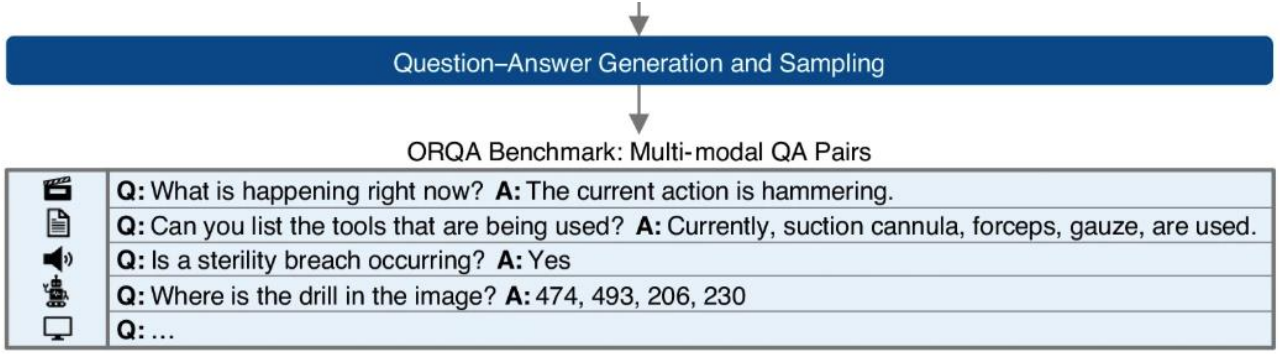


- CAIT moving more and more towards understanding of the context, patient history, and treatment properties
- Essential skill is the collaboration with medical experts (and patients) and workflow embedding

Surgical Data Science / Workflow Modelling

Surgical Data Science (SDS)

What is happening in a procedure — and what happens next?



Foundation Model Training
Özsoy et al. · 2026

Technical focus

Surgical phase recognition, instrument detection, surgical scene graphs.

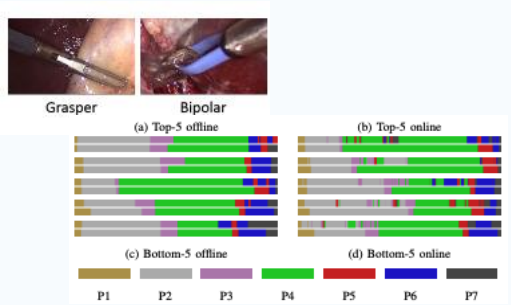
The hard part

Temporal context, reliable annotations, workflow variation and deployment.

Surgical Data Science

2017 FOUNDATION

EndoNet

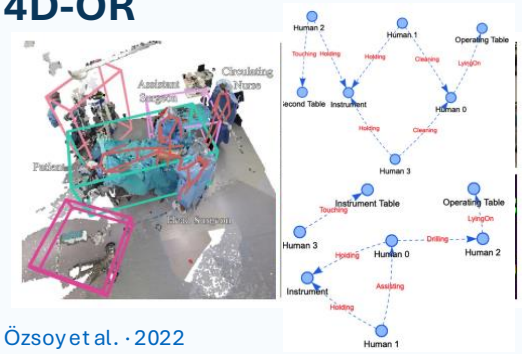


Twinanda et al. · 2017

Joint surgical tool and workflow recognition from video. Phase recognition that uses a convolutional neural network (CNN).

2022 RECENT

4D-OR

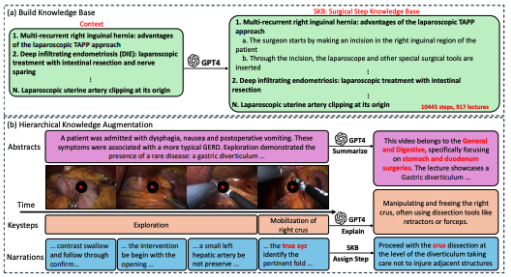


Özsoy et al. · 2022

From individual detections to relational OR understanding.

2025 RECENT

SurgVLP



Yuan et al. · 2025

Use narrated videos to learn open-vocabulary surgical semantics.

Camera-Based Surgery, Navigation & Planning

Camera-Based Surgery & Navigation

How does an image become useful guidance in the patient's space?



Image: B.Braun



Image: Brainlab

Technical focus

Medical imaging, tracking, spatial registration, planning and navigation.

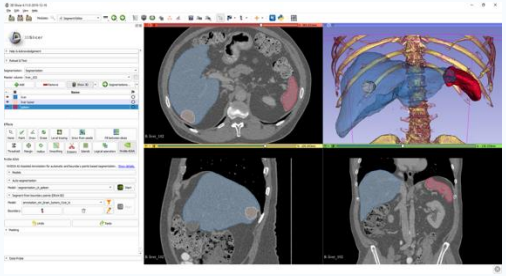
The hard part

Spatial accuracy, movement, anatomical deformation and uncertainty.

Medical image computing & planning

2012 FOUNDATION

3D Slicer

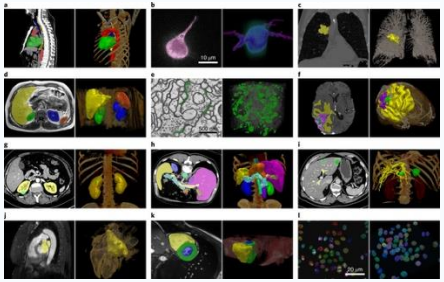


Fedorov et al. · 2012

Extensible medical-image computing, not a single algorithm; Segmentation, robot-assisted interventions, tracking, guidance… https://www.slicer.org/

2021 RECENT

V-Net & nnU-Net

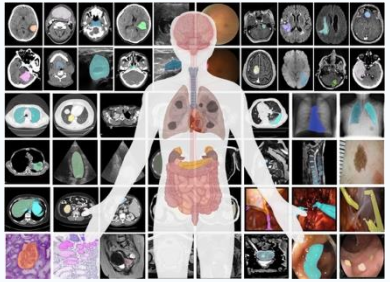


Milletari et al., 2016; Isensee et al. · 2021

V-Net is a volumetric CNN for 3d image segmentation.
nnU-Net is a deep learning-based segmentation method that automatically configures itself

2024 RECENT

MedSAM



Ma et al. · 2024

A promptable medical segmentation foundation model

Navigation, registration & reconstruction

1992 FOUNDATION

ICP

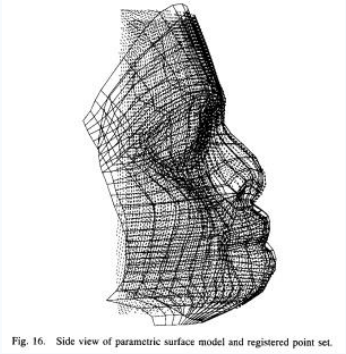


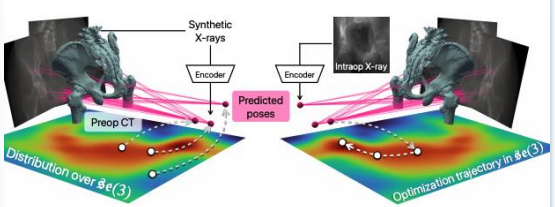
Fig. 16. Side view of parametric surface model and registered point set.

Besl & McKay · 1992

Iterative Closest Point. One of the core algorithms of rigid registration. Enables to match point-clouds.

2024 RECENT

DiffPose

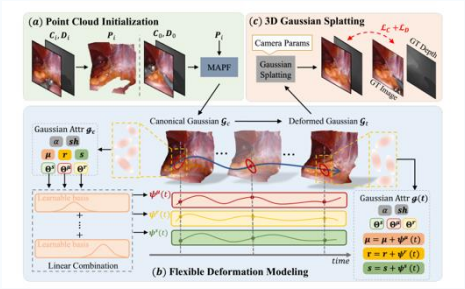


Gopalakrishnan, Dey & Golland · 2024

Differentiable physics enables patient-specific image-to-image registration

2024 RECENT

Deform3DGS



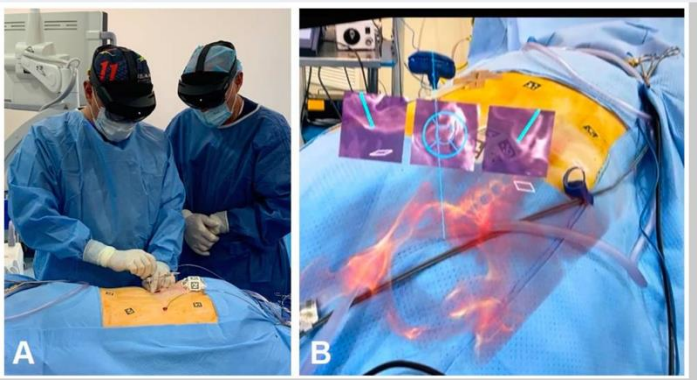
Yang et al. · 2024

Dynamic reconstruction despite deformable tissue

Advanced Displays / Medical XR

Advanced Displays & Medical XR

What should clinicians see, hear or feel — and when?



De Jesus Encarnacion Ramirez et al., , 2024



Martin-Gomez et al., 2020

Technical focus

Augmented reality, visual guidance, auditory and haptic feedback, multimodal interaction.

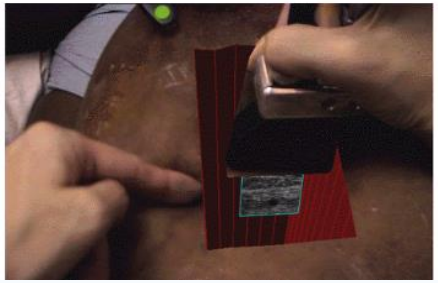
The hard part

Cognitive load, occlusion, latency, sterility, ergonomics and safety.

Medical AR & multimodal guidance

1996 FOUNDATION

AR ultrasound biopsy

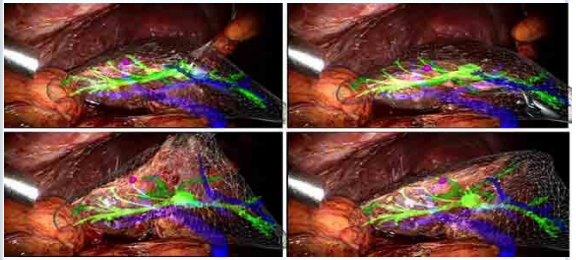


State et al. · 1996

Early real-time in-situ medical visualization

2015 RECENT

Markerless surgical AR

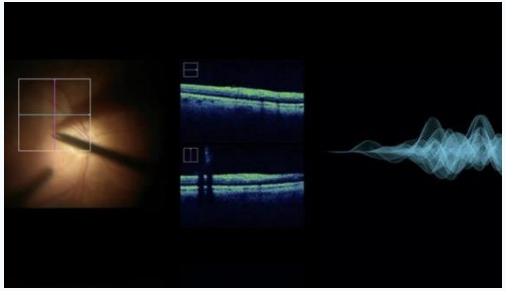


Hachoune et al. · 2015

AR for deformable liver surgery

2025 RECENT

From tissue to sound



Matinfar et al. · 2025

Mapping tissue properties to sonification output

Surgical Robotics

Surgical Robotics

Which movements should robots execute — and with what autonomy?



Technical focus

Teleoperation, constrained motion, robotic ultrasound, learning-based robotic assistance.

The hard part

Reliable sensing, soft tissue motion, haptics, safety and validating autonomy.

Surgical Robotics

1994 FOUNDATION

ROBODOC

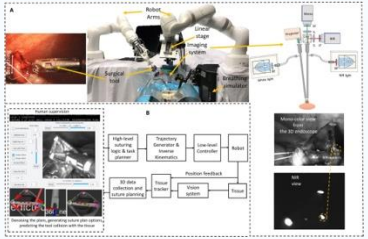


Taylor et al. · 1994
Image: insider.si.edu

Closed-loop image-directed robot execution of a surgical plan

2022 RECENT

STAR autonomous suturing



Saeidi et al. · 2022

Planning and autonomous manipulation of soft tissue

2025 RECENT

SRT-H



Kim et al. · 2025

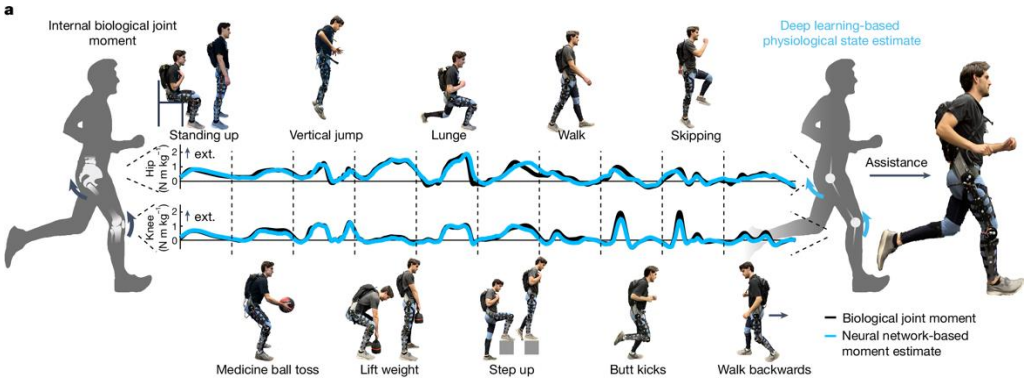
Language planning + learned motor skills in the loop

Human-Centered Computing and Extended Reality

Computer-Aided Therapy & Rehabilitation

Computer-Aided Therapy & Rehabilitation

Can we measure and adapt therapy beyond a single clinic visit?



Molinaro et al. · 2024

Technical focus

Wearable and video-based movement sensing, XR, adaptive therapy and assistive robots.

The hard part

Patient variability, long-term adherence, meaningful outcomes and personalization.

Human-Centered Computing and Extended Reality

Computer-Aided Therapy & Rehabilitation

1998 FOUNDATION

MIT-MANUS

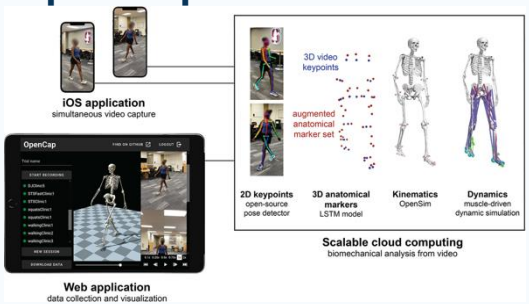


Krebs, Hogan, Aisen & Volpe · 1998

Quantified robot-mediated upper-limb motor training

2023 RECENT / INFLUENTIAL

OpenCap

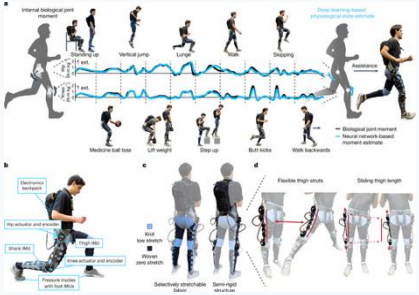


Uhlrich et al. · 2023

Accessible biomechanical assessment outside the motion-capture laboratory

2024 RECENT / INFLUENTIAL

Task-agnostic exoskeleton



Molinaro et al. · 2024

Learn biological moments instead of recognizing predefined gait modes

Human-Centered Computing and Extended Reality

Course dates, independent projects, & assessments

Course Dates

Course schedule · six theory sessions

Tuesdays · 16:00–18:00 · Room 5613.03.010, Garching

13 Oct	16:00–18:00	S01	**Kick-off, introduction, project pitches & voting**	Roth / Navab
20 Oct	16:00–18:00	S02	Surgical Data Science / Workflow Modelling	Navab, Killeen, Schieber
27 Oct	16:00–18:00	S03	Camera-Based Surgery, Navigation & Planning	Fehrentz, Theelke, Roth
03 Nov	16:00–18:00	S04	Advanced Displays / Medical XR	Roth, Kleinbeck, Matinfar, Ruozzi
10 Nov	16:00–18:00	S05	Surgical Robotics	Mukhashev, Bi
17 Nov	16:00–18:00	S06	Computer-Aided Therapy & Rehabilitation	Schaack, Roth

Project teams begin work and meet mentors alongside the lecture programme.

Course schedule · November and December

Fixed presentations vs. self-directed project windows

After matching	By agreement	KICK-OFF	**Individual team–mentor project kick-offs**
24 Nov	16:00–18:00	EXAM	**Intermediate oral presentations**
01 Dec	16:00–18:00*	PROJECT	Project work / mentor meetings by arrangement
08 Dec	16:00–18:00*	PROJECT	Project work / mentor meetings by arrangement
15 Dec	16:00–18:00*	PROJECT	Project work / mentor meetings by arrangement
22 Dec	—	BREAK	Christmas holidays — no session
29 Dec	—	BREAK	Christmas holidays — no session

**Regular course slot; no fixed plenary topic listed. Mentor meetings are individually arranged.*

Course schedule · January and February

Project work, self-organized mentor meetings and final examination

05 Jan	—	BREAK	Christmas holidays — no session
12 Jan	16:00–18:00*	PROJECT	Project work / mentor meetings by arrangement
19 Jan	16:00–18:00*	PROJECT	Project work / mentor meetings by arrangement
26 Jan	16:00–18:00*	PROJECT	Project work / final preparations
02 Feb	16:00–18:00	EXAM	**Final oral presentations and demonstrations**

**Course slot only; meetings with mentors take place at times agreed by each group.*

Project exposés and questions

Projects Exposés

- Listen to the pitches
- Make notes of your favorite projects and priorization
- Vote online in **Moodle** – you have 5 minutes after the presentations ☺
- Select priority projects, comment if you have a team member(s) you want to work with
- Not everyone will get the priority, but all projects are exciting ☺

Surgical Helmet

Mentors: Constantin Kleinbeck (constantin.kleinbeck@tum.de), Sasan, Veronica
Problem: Conventional surgical guidance relies on external screens, diverting attention from the task. Alternative, intuitive feedback methods could improve instrument alignment.
Background and Motivation: Helmet-integrated LEDs and auditory feedback offer a lightweight approach to spatial guidance without requiring a conventional AR display.
Goal: Develop a helmet-based guidance system for aligning a tracked probe with a target pose.
Proposed Solution:
Use OptiTrack to track probe pose and provide four feedback conditions:

Baseline: On-screen guidance, **Visual:** Helmet-integrated LEDs,
Auditory: Sonification of alignment errors, **Audiovisual:** Combined visual and auditory cues

Objectives:

- Integrate LEDs, speakers, and a microcontroller into existing surgical helmet.
- Design and test intuitive visual and auditory cues.
- Provide insights into alignment accuracy, completion time, and usability.

Tasks

- Design and build the helmet interface.
- Implement visual and auditory feedback.
- Test the system, conduct a user study and analyze results.
- Document findings and present the prototype.





1. AmbiGlasses



2. Stryker Surgical Helmet



3. 1-D Eyewear Prototype

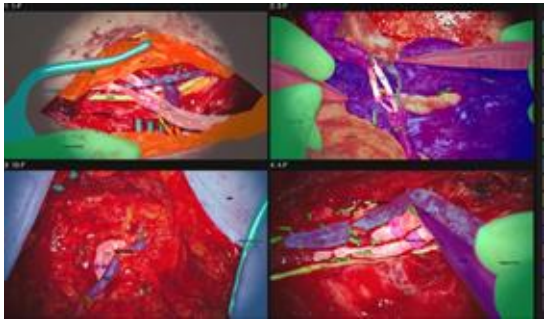
[1] B. Poppinga, N. Henze, J. Fortmann, W. Heuten, and S. Boll, "AmbiGlasses — Information in the Periphery of the Visual Field," in Mensch & Computer 2012: 12. fachübergreifende Konferenz für interaktive und kooperative Medien. interaktiv informiert – allgegenwärtig und allumfassend!?, 2012
[2] Image: https://www.stryker.com/de/de/orthopaedic-instruments/products/flyte-personal-protection-system.html
[3] A. Olwal and B. Kress, "1D eyewear: peripheral, hidden LEDs and near-eye holographic displays for unobtrusive augmentation," in Proceedings of the 2018 ACM International Symposium on Wearable Computers, Singapore Singapore: ACM, Oct. 2018
[4] J. Stewart and M. Billinghurst, "A wearable navigation display can improve attentiveness to the surgical field," Int J CARS, Jun. 2016

Human-Centered Computing and Extended Reality

Leveraging Routinely Post-Anastomosis Videos for Data-Efficient Microsurgical Anatomy Segmentation

- **Mentors:** Luisa (luisa.theelke@tum.de), Sinan (LMU) , Irene (LMU)
- **Problem:** Annotated microsurgical video data are scarce, and segmenting vessels, nerves, and instruments is challenging in complex intraoperative scenes.
- **Background and Motivation:** Routine post-anastomosis microscope videos provide controlled views of the surgical scene and are already densely annotated by medical experts. This project investigates whether these recordings can serve as an efficient source of supervision for more complex full-surgery scenes.
- **Goal:** Develop and evaluate segmentation approaches that learn from annotated post-anastomosis videos and generalize to full microsurgical procedures
- **Proposed Solution:** Adapt pretrained segmentation foundation models using annotated documentation videos and evaluate generalization to full-surgery scenes, including domain-adaptation strategies.
- **Tasks and Objectives:**
    - Investigate and compare suitable approaches.
    - Quantify the domain gap between documentation and full-surgery videos.
    - Explore domain-generalization/ domain-adaptation strategies
    - Evaluate data efficiency and the benefit of limited target-domain supervision.
    - Analyze failure cases and visualize predictions as overlays on the original videos.



Kamtam, D. N., Shrager, J. B., Malla, S. D., Wang, X., Lin, N., Cardona, J. J., ... & Hu, C. (2025). A fine-tuned foundational model SurgiSAM2 for surgical video anatomy segmentation and detection. Scientific Reports, 15(1), 35961.
Yuan, C., Jiang, J., Yang, K., Wu, L., Wang, R., Meng, Z., ... & Ban, Y. (2026). Systematic evaluation and guidelines for segment anything model in surgical video analysis. npj Digital Surgery, 1(1), 2.Tang, M., et al. (2024). Assessment of changes in vessel area during needle manipulation in microvascular anastomosis using a deep learning-based semantic segmentation algorithm. Neurosurgical Review, 47, 200.
Jaspers, T. J. M., et al. (2026). Scaling up self-supervised learning for improved surgical foundation models. Medical Image Analysis, 108, 103873.
Yang, S., et al. (2026). Large-scale self-supervised video foundation model for intelligent surgery. npj Digital Medicine, 9, 220.

Human-Centered Computing and Extended Reality

World Models for Movement Prediction in the OR

Mentors: Hannah Schieber (hannah.schieber@tum.de), Benjamin Killeen

Problem: Existing surgical video models focus on recognition or scene-level prediction. The goal is to learn structured OR representations from single- or multi-camera videos and predict future surgical activities and room dynamics.

Background and Motivation: World models learn environmental dynamics to predict future states and simulate possible outcomes.

Goal: Develop a world model that learns a structured OR representations outside of endoscopic videos, predicts future scene states, and generalizes to unseen surgical scenarios while maintaining semantic and visual consistency.

Proposed Solution: Encode surgical videos into latent OR states and predict future states using learned dynamics, with surgical phase labels as semantic supervision. Train and evaluate on available datasets, measuring phase consistency, temporal coherence, visual quality, and generalization.

Objectives:

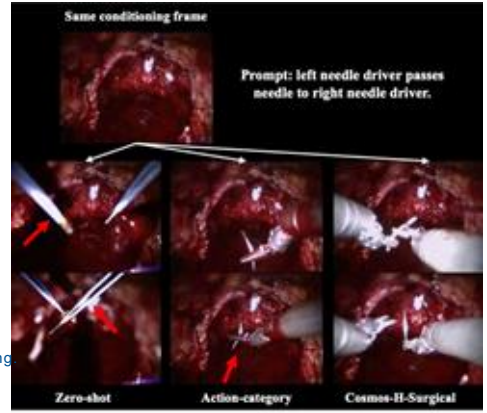
- Learn predictive OR scene representations using surgical video and phase-label supervision.
- Evaluate future-state predictions for semantic consistency, temporal coherence, and visual quality on unseen sequences.

Tasks

- *Data Preparation*: Prepare surgical data for model training and evaluation.
- World-State Representation: Design a latent representation of OR scenes.
- *Model Development*: Implement and train a world model to predict future latent states.
- *Future-State Generation:* Generate plausible future OR scenes from observed video clips.
- *Evaluation*: Assess prediction accuracy, consistency, temporal coherence, and generalization.







Özsoy, E., Örnek, E. P., Eck, U., Czempiel, T., Tombari, F., & Navab, N. (2022, September). 4d-or: Semantic scene graphs for or domain modeling. MICCAI
Özsoy, E., Mamur, A., Tristram, F., Pellegrini, C., Wysocki, M., Busam, B., & Navab, N. (2026). Egoexor: An ego-exo-centric operating room dataset for surgical activity understanding. *Neurips*
Zhu, H., Liu, H., Zhao, Y., Ye, T., Chen, J., Yu, J., ... & Xie, E. (2026). Sana-wm: Efficient minute-scale world modeling with hybrid linear diffusion transformer. *arXiv*
Chen, J., Zhao, Y., Yu, J., Chu, R., Chen, J., Yang, S., ... & Xie, E. (2026, April). Sana-video: Efficient video generation with block linear diffusion transformer. ICLR
He, Y., Guo, P., Xu, M., Li, Z., Myronenko, A., Imans, D., ... & Xu, D. (2025). Cosmos-H-Surgical: Learning Surgical Robot Policies from Videos via World Modeling. *arXiv*

Human-Centered Computing and Extended Reality

Multimodal Force Feedback for Teleoperated Bone Cutting: Which Sensory Channels Work Best?

Mentor: Dimash
dimash.mukashev@tum.de

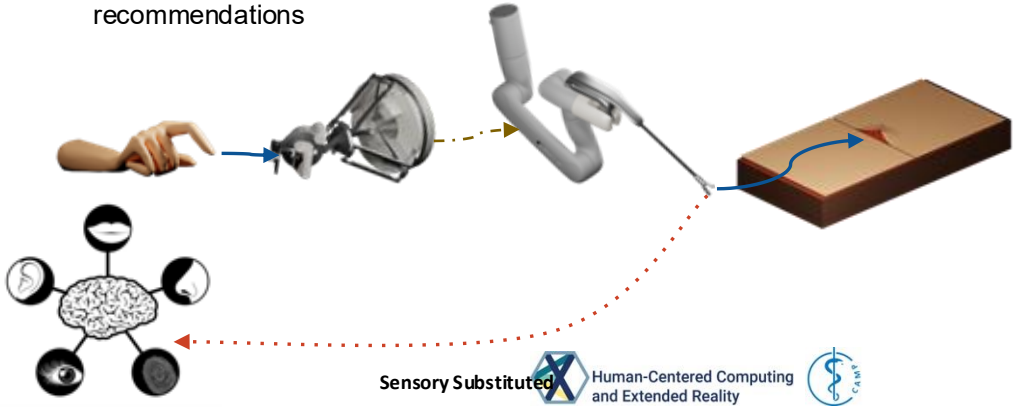


- **Topic Area:** Human-in-the-loop teleoperation and haptics for surgical robotics (embodiment, sensory substitution, mock-OR user studies, ROS-based testbeds)
- **Problem:** In teleoperated orthopedic surgery, the surgeon is physically separated from the patient. Simply mirroring the large, high-frequency tool–bone forces to the control hand overloads the operator, destabilizes the control loop, and masks the robot's guidance cues (sensory ambiguity).
- **Background and Motivation:** Robotic systems like Mako improve cutting precision but keep the surgeon at the bedside, hands-on. Moving to teleoperation requires rerouting interaction forces to other sensory channels (visual, auditory, tactile) — but each single channel has known limits (visual overload, auditory masking, low tactile bandwidth). Which combinations work, and for which task phases, is an open question.
- **Goal:** Systematically compare feedback modality combinations (visual / auditory / vibrotactile / kinesthetic) in a teleoperated cutting task, measuring task performance and cognitive load.

- **Proposed Solution:** Extend an existing ROS-based teleoperation testbed with a multimodal feedback pipeline; run a within-subject user study on standardized silicone/bone phantoms;

Tasks:

- Familiarize with the teleoperation testbed (Franka, Sigma.7, ROS)
- Implement 2–3 feedback mappings (e.g., sonification, vibrotactile display, visual overlay)
- Design and fabricate standardized phantom task (with lab support)
- Conduct a pilot user study (N≈12–15) comparing feedback conditions
- Analyze performance, force consistency, and workload; report design recommendations

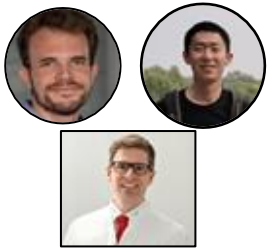


Okamura, Allison M. "Haptic feedback in robot-assisted minimally invasive surgery." Current opinion in urology 19.1 (2009): 102.
Chinello, Francesco, et al. "A modular wearable finger interface for cutaneous and kinesthetic interaction: control and evaluation." IEEE Transactions on Industrial Electronics 67.1 (2019): 706-716.
Saracino, Arianna, et al. "Haptic feedback in the da Vinci Research Kit (dVRK): A user study based on grasping, palpation, and incision tasks." The International Journal of Medical Robotics and Computer Assisted Surgery 15.4 (2019): e1999.

Surgical Tattoos for Knee Arthroplasty

- **Topic Area:** Computer vision, fiducial markers
- **Problem:** Robotic-assisted knee surgeries (e.g., MAKO) are more precise than manual surgeries, but require dynamic tracking of the leg bones via optical markers which are pinned into the bones. This is necessary for the robotic assistance, but harms the bone.
- **Background and Motivation:** Less invasive would be visual markers that are applied to the skin, such as tattoos. However, soft tissue displacement occurs here, and the positions of the marker relative to the bone is not rigid. This makes it tricky to estimate the bone axes.
- **Goal:** Experimental evaluation of a non-invasive skin-based tracking method for tibia and femur axes
- **Proposed Solution:** Explore and compare skin-based marker approaches to ground truth data (e.g., using OptiTrack)
- **Tasks and Objectives:** Using an artificial knee, create a prototype which compares skin-based marker precision to ground truth femur and tibia axes.

- **Mentors**: Julian, Shiyu, Dr. med Dirk Müller



julian.kreimeier@tum.de
shiyu.li@tum.de





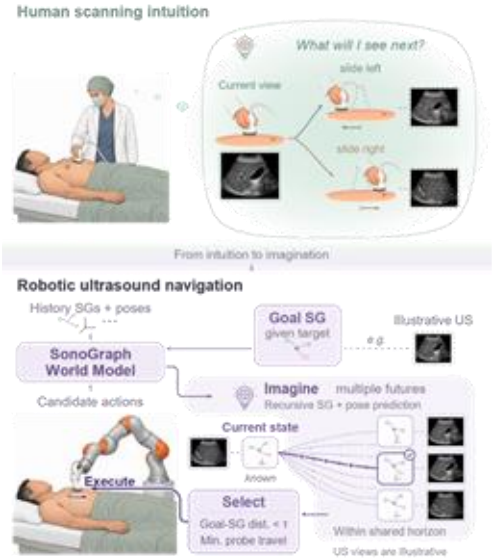


Coburn, K., Velhal, M., Goenezen, S., & Liang, H. (2023). Digital image correlation through surface patterning of skins to monitor knee motion. Biotribology, 35, 100251.
Zamora, R., Punt, S. E., Christman-Skieller, C., Yildirim, C., Shapton, J. C., & Conrad 3rd, E. U. (2019). Are skin fiducials comparable to bone fiducials for registration when planning navigation-assisted musculoskeletal tumor resections in a cadaveric simulated tumor model?. Clinical Orthopaedics and Related Research, 477(12), 2692.
Tsukada, S., Ogawa, H., Nishino, M., Kurosaka, K., & Hirasawa, N. (2019). Augmented reality-based navigation system applied to tibial bone resection in total knee arthroplasty. Journal of experimental orthopaedics, 6(1), 44
Tsukada, S., Ogawa, H., Nishino, M., Kurosaka, K., & Hirasawa, N. (2021). Augmented reality-assisted femoral bone resection in total knee arthroplasty. JBJS Open Access, 6(3), e21-00001.
Ancillao, A., Aertbeliën, E., & De Schutter, J. (2021). Effect of the soft tissue artifact on marker measurements and on the calculation of the helical axis of the knee during a gait cycle: A study on the CAMS-Knee data set. Human Movement Science, 80, 102866.
Rath, B., Huber, T., Kamper, M., Beckmann, J., Hirschmann, M. T., Hammer, N., & Ullmann, D. (2026). Feasibility and accuracy of intraoperative 3D surface scanning to assess femoral component alignment relative to the native distal femoral articular surface in total knee arthroplasty. Knee Surgery, Sports Traumatology, Arthroscopy.
Agarwal, B. B. (2009). Tattoo: a navigational tool, a lighthouse for biomolecular level precision in surgical endoscopy. Newton's apple or Eve's?. Surgical endoscopy, 23(5), 1166-1167.

World Model for Robotic Ultrasound Navigation

- **Mentors:** Yuan Bi and Xuesong Li
- **Topic Area:** Robotic ultrasound World models
- **Problem:** Robots move the probe without anticipating how the anatomy will change.
- **Background & Motivation:** Our SonoGraph-WM plans by imagining future scene graphs. Label maps keep richer anatomy.
- **Goal:** Extend SonoGraph-WM to predict future label maps and probe poses.
- **Proposed Solution:** Train on CT-derived label-map–pose sequences; imagine candidate paths and pick the shortest one reaching the goal.
- **Contact:** yuan.bi@tum.de & xuesong.li@tum.de
- **Tasks and Objectives**
  - 1. Generate label-map–pose data fromCT
  - 2. Train the label-map world model
  - 3. Plan and navigate in simulation
  - 4. Bonus: robot–phantom test
- **5 Related Works**
  - 1. Li et al., SonoGraph-WM, arXiv 2026
  - 2. Li et al., US scene graphs, MICCAI 2025
  - 3. Fan et al., US world model, arXiv 2026
  - 4. Yue et al., EchoWorld, CVPR 2025
  - 5. Hafner et al., PlaNet, ICML 2019

Teaser Image



SonoGraph-WM; this project replaces scene graphs with label maps.

Human-Centered Computing and Extended Reality

“Towards imitation learning for exoskeletons” - Developing a knee angle forecast engine





Mentors: Victor Schaack, Daniel Homm, Christina Valle

Topic Area: Movement analysis, prediction, robotic simulation

Background and Motivation: Increasing usability for exoskeletons requires the system to predict the wearers intent. This can be done by imitating normal human motion.

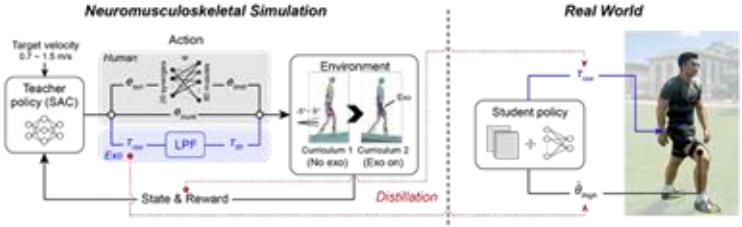
Problem: Current models are highly susceptible to different sensor placements and really heavily on EMGs. This increases intra-subject variability to too high a degree to inform a robotic control policy.

Goal: Develop a pipeline, in Isaac Lab or Mujoco, that predict knee angles using 2/4 arbitrarily placed IMUs per leg segment.

Proposed Solution: Teacher student policy, with the teacher policy having access to EMG data, optical tracking data and IMU data, while the student policy is distilled to down to only IMU data, just like deployed on the exoskeleton.

Tasks and Objectives:

1. (Confer with medical experts which movements are most relevant for rehabilitation use-case).
2. Dataset suitability (rehab-movement included, sensors, frequencies)
3. Synthetic data augmentation to suit exoskeleton IMU placement
4. Teacher-student distillation (EMG + optical tracking + IMU → IMU only)
5. ROS bridge that communicates knee angles for later robotic use.



Park, I., Song, C., Kang, I. (2026). Learning Hip Exoskeleton Control Policy via Predictive Neuromusculoskeletal Simulation. arXiv preprint arXiv:2603.04166.

Saoud, L. S., & Hussain, I. (2023). TempoNet: Empowering Long-Term Knee Joint Angle Prediction with Dynamic Temporal Attention in Exoskeleton Control. In 2023 IEEE-RAS 22nd International Conference on Humanoid Robots (Humanoids)

Coker, J., Chen, H., Schall, M. C., Jr., Gallagher, S., & Zabala, M. (2021). EMG and Joint Angle-Based Machine Learning to Predict Future Joint Angles at the Knee. Sensors, 21(11), 3622.

Sharifi Renani, M., Eustace, A. M., Myers, C. A., & Clary, C. W. (2021). The Use of Synthetic IMU Signals in the Training of Deep Learning Models Significantly Improves the Accuracy of Joint Kinematic Predictions. Sensors, 21(17), 5876.

Zaroug, A., Garofolini, A., Lai, D. T. H., Mudie, K., & Begg, R. (2021). Prediction of gait trajectories based on the Long Short Term Memory neural networks. PLOS ONE, 16(8), e0255597.

Su, B., & Gutierrez-Farewik, E. M. (2020). Gait Trajectory and Gait Phase Prediction Based on an LSTM Network. Sensors, 20(24), 7127

Human-Centered Computing and Extended Reality

Project preference voting

Please go to the moodle course

… and vote ☺

- We will announce the groups by Thursday this week.

Next Steps & Closing

Project-supervision is self-driven

- Each project team meets its assigned mentors separately – not in a class-wide supervision session.
- Contact the mentors proactively, arrange kick-off and plan regular follow-up meetings.
- Come prepared: share progress, decisions, test results and questions with a slide-deck.
- Work independently between meetings; report blockers early and track individual contributions.

- Mentors advise – students lead
  - Close interaction is essential, but organizing it and driving the progress is your responsibility.

Your first meeting: the kick-off

After the project allocations – date and time agreed with the assigned mentors

1. Contact your mentors: The student team requests a kick-off appointment.
2. Align on scope: Clarify the clinical problem, outputs, and responsibilities.
3. Check feasibility: Verify data/hardware, baseline, metrics, and approvals.
4. Agree to next actions: Set milestones, ownership, and the next check-in.

There is no preassigned date for these meetings: schedule them directly with your mentors.

Perspectives in your projects

1	**Clinical need**	Who benefits and why?
2	**Technical task**	What is to be built?
3	**Starting point**	What data/hardware exists?
4	**Evaluation**	What would count as evidence?
5	**Feasibility**	What can be delivered this semester?

NEXT Week

Surgical Data Science / Workflow Modelling

Prof. Navab, Dr. Killeen, Hannah Schieber

Thank you!

Questions?